4 (a) (i) 1. stress = force / (cross-sectional) area B1 [1]
2. strain = extension / <u>original</u> length B1 [1]
3. Young modulus = stress / strain B1 [1]
(ratios must be clear in each answer)

(ii) *either* fluids cannot be deformed in one direction / cannot be stretched
or fluids can only have volume change
or no fixed shape B1 [1]

(b) *either* unless Δp is very large *or* 2.2×10^9 is a large number M1
ΔV is very small *or* $\Delta V/V$ is very small, (so 'incompressible') A1 [2]

(c) $\Delta p = h\rho g$
$1.01 \times 10^5 = h \times 1.08 \times 10^3 \times 9.81$ C1
$h = 9.53$ m C1
$\Delta h / h = 0.47 / 10$ *or* $0.47 / 9.53$
error = 4.7% *or* 4.9% *or* 5% A1 [3]

5 (a) (i) frequency: number of oscillations per unit time M1